PORTFOLIO PROJECT

Automated onboarding.

HR sends a message.
The right person gets it in their language.
Every step is logged. Nothing falls through.

253	20	0
tasks created	new hires	messages sent without approval

What this project demonstrates

1. **Building a complete automation system**
   4 automated flows, a PostgreSQL database, and a dedicated error safety net — all connected.

2. **Handling real HR complexity**
   Three hire types. Bilingual messages. Urgent vs. routine routing. A blocked hire logged before anything breaks.

3. **Validating that it actually works**
   253 tasks created. Bangla message confirmed in send queue. Nothing sent without approval. Errors caught and logged.

n8n · PostgreSQL · bilingual · HR automation

WHY THIS WAS BUILT

Manual onboarding breaks quietly.

HR was tracking onboarding tasks manually across three hire types. Things slipped. No one was alerted. No audit trail. Messages were English-only.

No reminders — Tasks went overdue with no alert to anyone.

No escalation — A week-late task had no defined owner.

English-only — Factory and field workers missed instructions.

Nothing logged — No record of what was sent, blocked, or approved.

Reminders

Tasks went overdue.
No one was notified.
HR found out too late.

Escalation

Urgent and routine tasks looked the same.
No approval path existed.

Language

Messages went out in English only.
Factory workers missed them.

Audit trail

No record of errors, blocked hires, or what was approved.

THE SYSTEM

Four flows.
One job each.

Each flow has a single responsibility.

A dedicated safety net catches any errors.

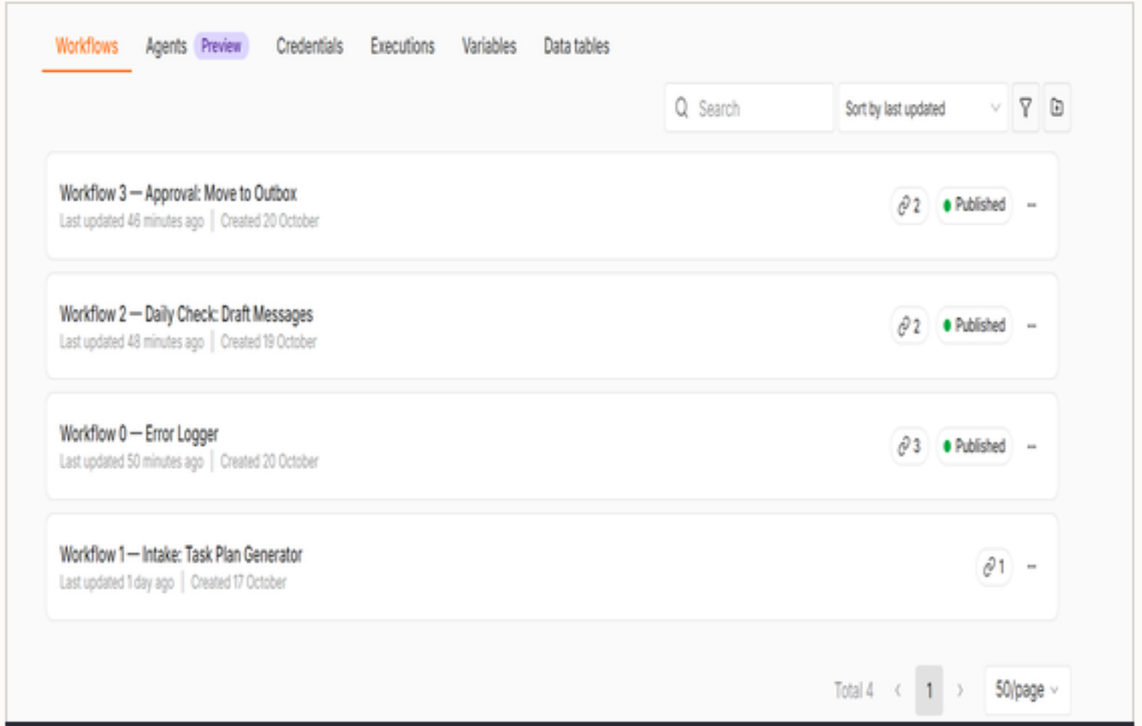
0 **Setup**
HR triggers once per new hire. Builds the task plan. Flags missing information.

1 **Daily check**
Runs at 8:00 AM. Finds overdue tasks. Routes to approval or sends directly.

2 **Approval**
Checks every 15 minutes. Moves approved messages to the send queue. No duplicates.

3 **Safety net**
Catches any error from the other flows. Logs it automatically for HR to review.

`new hire → task plan → daily check → approval → send queue`



n8n — all 4 flows published and active

Database — 6 tables

Hires
new hire records

Tasks
per-hire task instances

Send queue
messages ready to send

Task templates
master task lists

Pending approval
urgent messages for HR

Event log
errors and blocked hires

HOW IT RUNS

HR triggers it.
It builds the plan.

When a new hire joins, HR runs this flow once. It checks for blockers first, then builds the full task plan automatically.

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1 Check for blockers

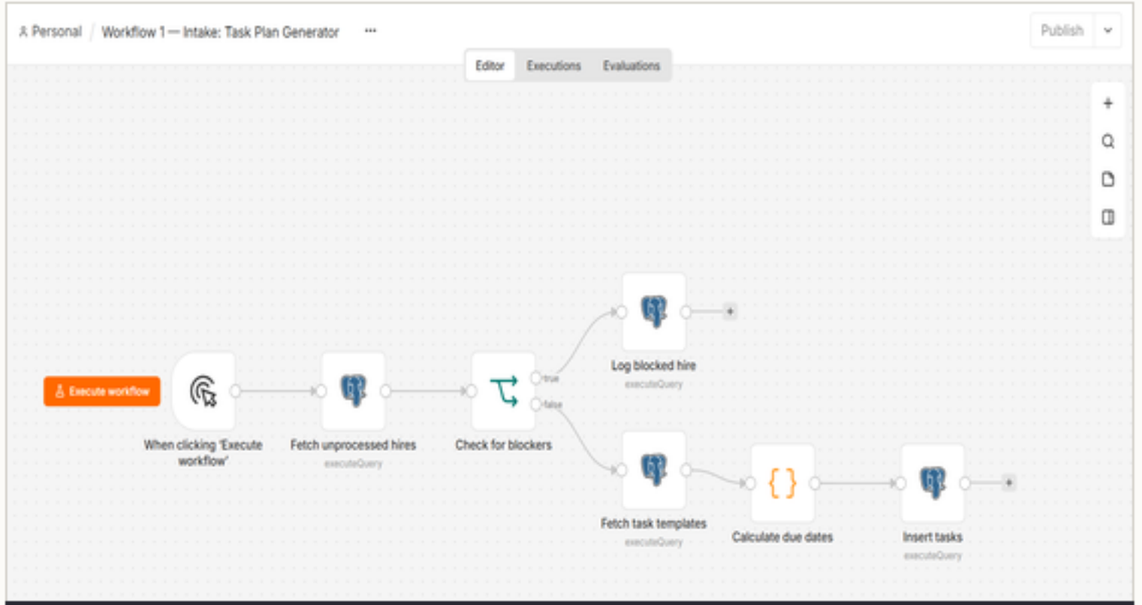
If a head office hire has no manager assigned, log it and stop. No tasks are created.

2 Build the task plan

Pull the right task list for this hire type. Calculate due dates. Save all tasks.

3 Mark as done

Flag the hire as set up. 19 of 20 hires passed. 253 tasks created in total.



New hire setup flow — 6 steps

253	19	1
tasks created	hires set up	hire blocked

What triggers a block:

Head office hire with no manager assigned → block logged, no tasks created

EVERY MORNING

Finds overdue tasks.
Routes them correctly.

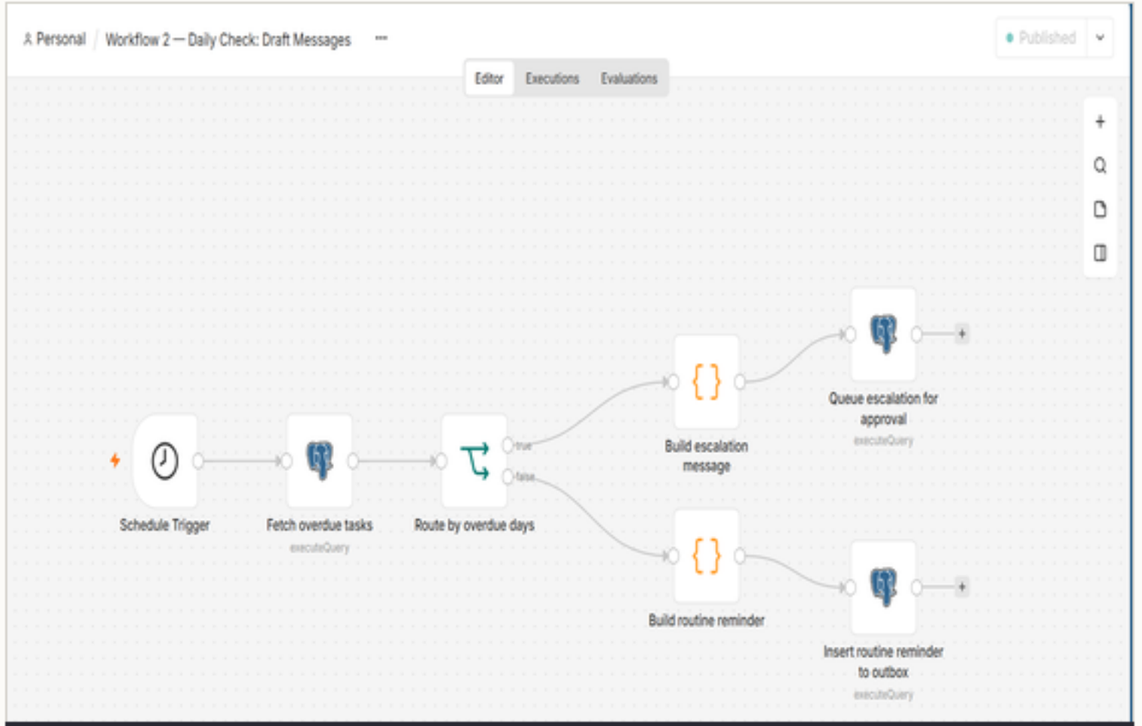
Runs at 8:00 AM every day. Finds tasks that are overdue.
Routes each one based on how late it is.
Builds the message in the hire's language.

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3 or more days overdue
Message goes to an approval queue.
HR reviews and approves before it's sent.

Under 3 days overdue
Message goes straight to the send queue.
No approval needed for routine reminders.

Language rule: message is built in the hire's preferred language — Ba nglish



Daily check flow — 7 steps — 8:00 AM schedule

pass Bilingual messages confirmed

pass No external API needed

pass Approval gate working

pass Errors caught automatically

THE GATE

Nothing goes out without sign-off.

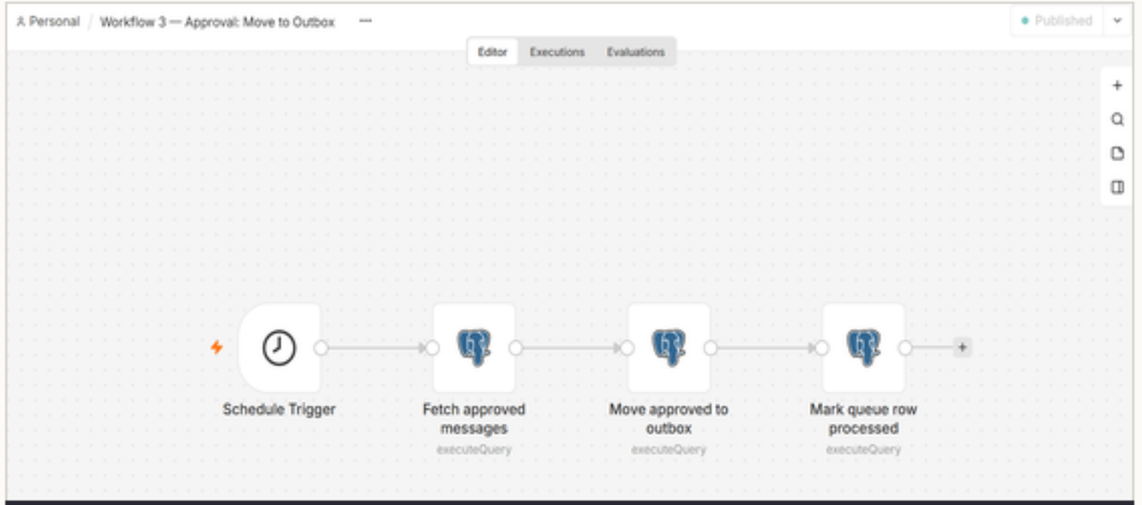
Checks every 15 minutes for messages HR has approved.
Moves them to the send queue. Marks each one done.
Safe to run again — it never sends the same message twice.

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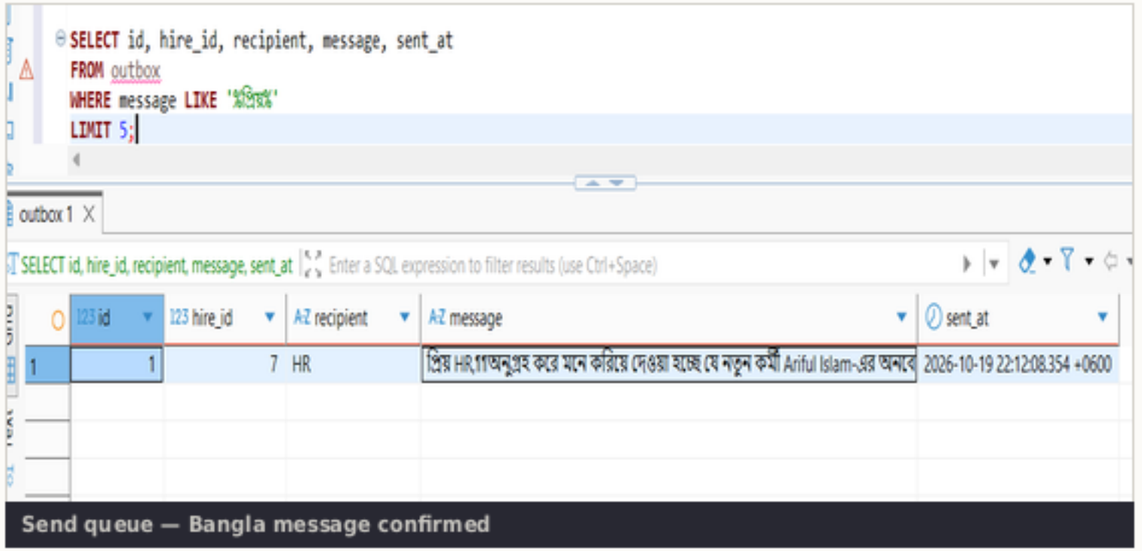
1. **Find approved messages**
   Look in the approval queue for messages
   HR has approved and not yet processed.

2. **Move to send queue**
   Copy the message across with hire details.
   It's now ready to send.

3. **Mark as done**
   Flag it as processed.
   It won't be moved again.



Approval flow — 4 steps



Send queue — Bangla message confirmed

THE RESULTS

Four checks.
Four passes.

pass

Nothing sent without approval

0 messages in the send queue that weren't approved by HR first.

pass

Bilingual messages confirmed

A hire with Bangla preference received a Bangla message in the send queue.

pass

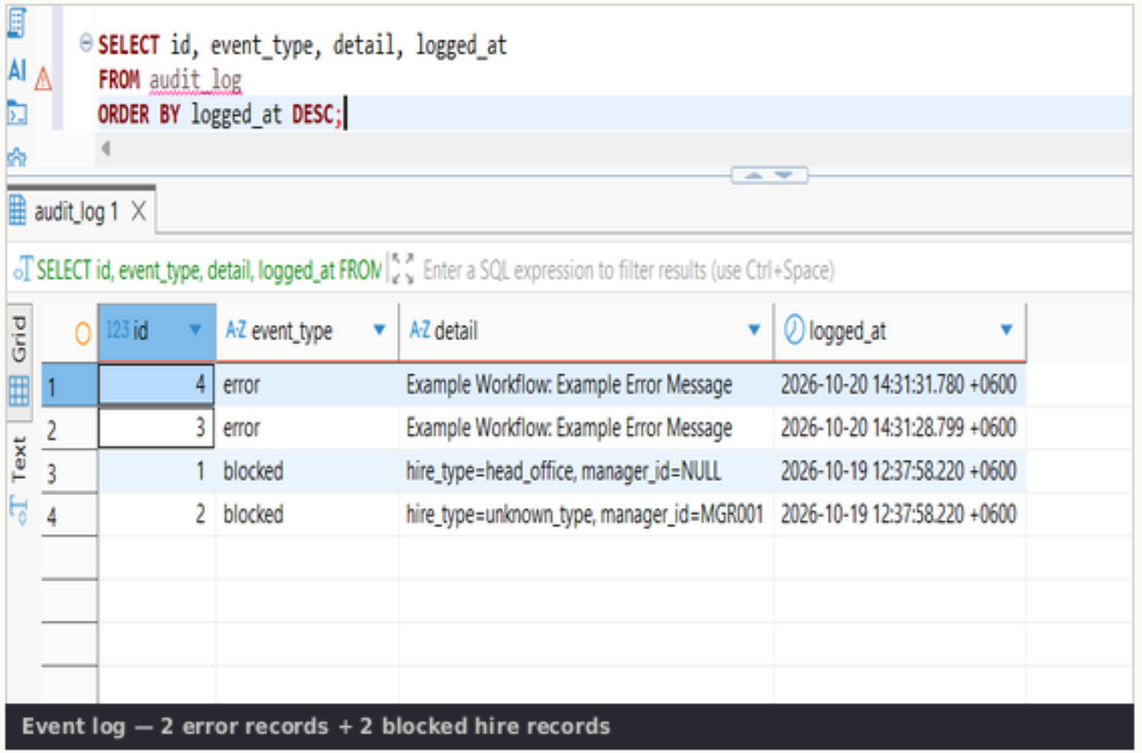
Blocked hire recorded

A head office hire with no manager was blocked and logged — no tasks created.

pass

Errors caught and logged

2 error events in the log. The safety net caught them automatically.



Event log — 2 error records + 2 blocked hire records

253 tasks created | **20** new hires | **4** flows | **2** languages | **0** unapproved